MATHEMATICS

# Applications and interpretation

Higher Level | Set D | Paper 2

## Challenge mock examination

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

## Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• All questions are compulsory. There is no optional section.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

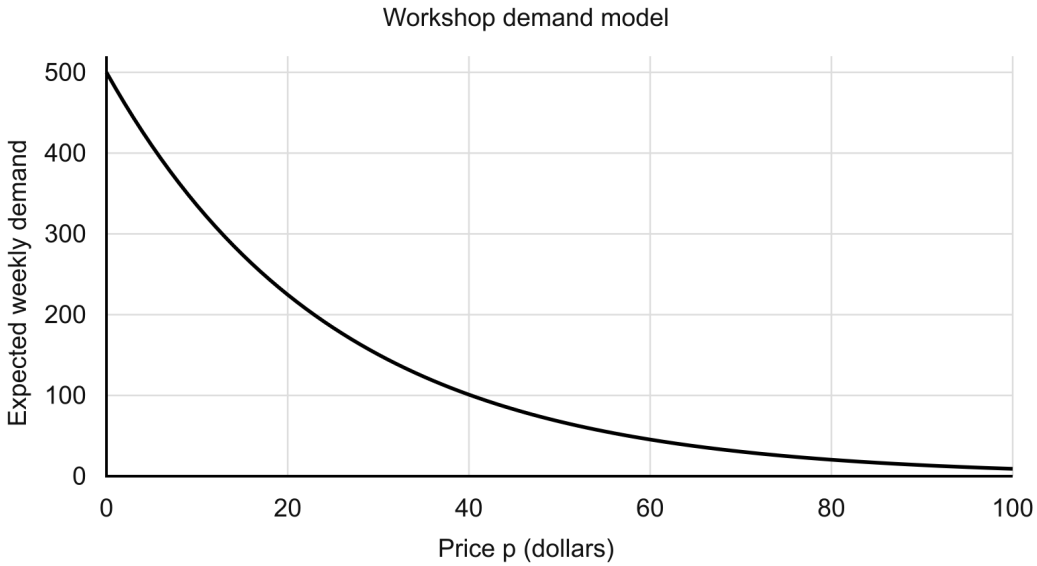
• The number in square brackets gives the maximum mark for a part.

Thriving Scholars original practice | 2026 challenge series

Question 1 | Question paper

## Question 1 [Maximum mark: 18]

A workshop models weekly demand for a product as $q(p) = 500e^{-0.04p}$, where $p$ is the selling price in dollars, $0 \leq p \leq 100$. Every demanded unit can initially be supplied. Each unit costs \$8 to make and fixed weekly costs are \$1200. The expected weekly profit is revenue minus all production and fixed costs.




(a) Write the profit function $\Pi(p)$ and find its derivative. [4]

(b) Find the price giving maximum expected profit, and the corresponding demand and profit. Justify the maximum on the stated domain. [3]

(c) Find the interval of prices giving non-negative expected profit. [4]

(d) Production capacity is reduced to 100 units per week. The workshop chooses a price at which the modelled demand does not exceed capacity. Find the price maximizing expected profit under this restriction, and its profit. [4]

Question 1 | Question paper

(e) If only fixed weekly costs increase, explain how the unconstrained profit-maximizing price and the set of profitable prices are affected. [3]

Question 2 | Question paper

## Question 2 [Maximum mark: 18]

A weather model has states sunny $(S)$ and rainy $(R)$. In either state, the probability of staying in that state on the next day is $0.75$. Day $0$ is sunny. A solar unit produces $10$ kWh on a sunny day and $3$ kWh on a rainy day. The Markov state for each day determines that day's production. Let $p_n = P(\text{day } n \text{ is sunny})$.

(a) Write the transition matrix using column state vectors ordered $(S, R)$, and show that $p_{n+1} = 0.5p_n + 0.25$. [3]

(b) By considering $p_n - 0.5$, obtain an explicit expression for $p_n$. [3]

(c) Find the steady-state distribution and explain why this chain has a unique long-run distribution. [2]

(d) Find the probability that exactly two of days $1, 2, 3$ are sunny. [4]

(e) Find the expected total energy produced over days $1$ to $10$, showing a geometric sum. [4]

(f) Explain why it would be incorrect to model the number of sunny days among days $1, 2, 3$ as $\text{Bin}(3, 0.5)$. [2]

## Question 3 [Maximum mark: 19]

An environmental scientist records independent pairs of water temperature $x$ in degrees Celsius and dissolved oxygen $y$ in mg per litre from eight randomly selected comparable sites. For the correlation test, assume the population of pairs is bivariate normal.

| $x$ | 12 | 14 | 16 | 18 | 20 | 22 | 24 | 26 |
|---|---|---|---|---|---|---|---|---|
| $y$ | 10.4 | 9.8 | 9.7 | 8.8 | 8.6 | 8.2 | 7.9 | 7.4 |

(a) Find Pearson's correlation coefficient and the least-squares regression line of $y$ on $x$. [4]

(b) At the $1\%$ significance level, test whether the population correlation is negative. State hypotheses and report an appropriate test statistic, degrees of freedom, $p$-value and conclusion. [5]

(c) Find Spearman's rank correlation coefficient and explain why it differs from Pearson's coefficient here. [3]

(d) Predict dissolved oxygen at $21^{\circ}\text{C}$ and find the residual for the observed site at $20^{\circ}\text{C}$. [3]

(e) A ninth recorded pair is $(40, 12)$. Find the new Pearson coefficient. Discuss whether the extra observation should automatically be deleted and whether the original correlation establishes a causal effect of temperature. [4]

Question 4 | Question paper

## Question 4 [Maximum mark: 19]

A transport researcher independently samples $60$ residents at random within each of three zones. Each resident reports one main travel mode. No resident appears twice. The observed counts are:

| Zone | Bus | Walk | Cycle |
|---|---|---|---|
| 1 | 30 | 20 | 10 |
| 2 | 18 | 25 | 17 |
| 3 | 12 | 15 | 33 |

The three zones have different population sizes; equal sample sizes were chosen for convenience.

(a) Under independence of zone and travel mode, find the expected-frequency table and check the expected-count condition. [3]

(b) Carry out a chi-square independence test at the $5\%$ significance level. State hypotheses, degrees of freedom, statistic, $p$-value and conclusion. [5]

(c) Identify the single cell making the largest contribution to the statistic. Calculate that contribution and interpret the direction of its deviation. [3]

(d) Zones 2 and 3 are combined into one category. Repeat the independence test for the resulting $2 \times 3$ table and state whether the conclusion changes. [4]

(e) Explain why the pooled sample proportion travelling by bus need not estimate the whole area's bus-use proportion accurately. Give a suitable adjustment if zone population sizes are known. [4]

Question 5 | Question paper

## Question 5 [Maximum mark: 18]

A damped displacement $x(t)$ satisfies the given equation $x'' + 3x' + 2x = 0$, with $x(0) = 4$ and $x'(0) = -2$. Displacement is in centimetres and time is in seconds. Let $v = x'$. The model is used only for $t \geq 0$.

(a) Write the model as a first-order matrix system for $(x, v)^T$, including its initial state. [3]

(b) Find the eigenvalues and corresponding eigenvectors of the system matrix. Classify the equilibrium at the origin. [4]

(c) Use the eigensolutions and initial state to find $x(t)$ and $v(t)$. [5]

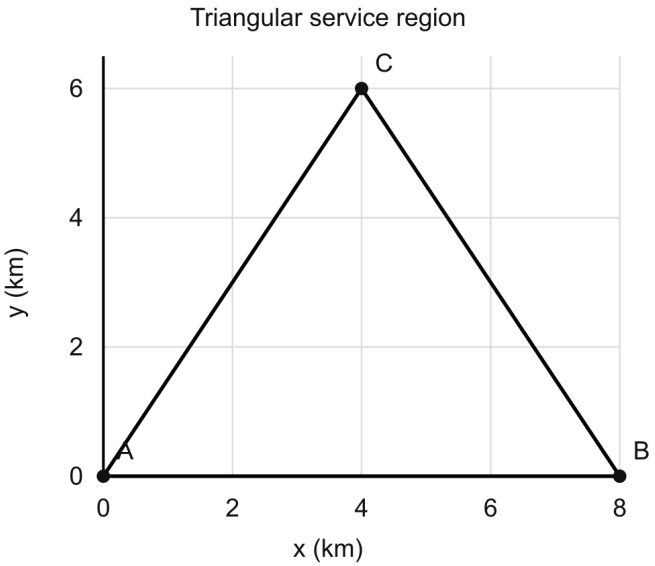
(d) Show that the displacement remains positive and decreases for all $t \geq 0$. Explain what the model predicts in the long term. [3]

(e) Use Euler's method on the coupled first-order system, with step size $0.2$ seconds, to estimate the state at $t = 0.4$. Compare it with the exact state. Both updates at a step must use the same old state. [3]

Question 6 | Question paper

## Question 6 [Maximum mark: 18]

Three emergency posts are at $A(0,0)$, $B(8,0)$ and $C(4,6)$, with coordinates in kilometres. The service region is exactly the closed triangular region $ABC$. Straight-line distance determines the nearest post. A transmitter is to be placed as far as possible from its nearest post while remaining in this triangle. For a convex polygonal cell, you may use the fact that the maximum distance from its site occurs at a cell vertex.




(a) Find the perpendicular bisectors of $AB$ and $AC$. [4]

(b) Find the common Voronoi vertex and its distance from each post. Check that it lies inside the triangle. [4]

(c) Find the area of the portion of the triangle allocated to post $A$. [4]

(d) Find the transmitter's optimal position and the maximum distance from its nearest post. Justify the global optimum by checking all types of cell vertex, including those on the triangle boundary. [4]

Question 6 | Question paper

(e) The transmitter must instead be on the base road $AB$. Find its best position and maximum nearest-post distance. [2]

## End of question paper

MATHEMATICS

# Applications and interpretation

Higher Level | Set D | Paper 2

## Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 | Worked solutions

# Question 1 [Maximum mark: 18]

## (a) [4 marks]

**M1** Revenue is $pq(p)$ and variable cost is $8q(p)$.

**A1** $\Pi(p) = 500(p-8)e^{-0.04p} - 1200$.

**M1** Differentiate using the product rule.

**A1** $\Pi\prime(p) = 500e^{-0.04p}\{1 - 0.04(p-8)\}$.

## (b) [3 marks]

**M1** $1 - 0.04(p-8) = 0$ gives $p = 33$.

**R1** The derivative changes from positive to negative there, so it gives the global maximum on $[0, 100]$.

**A1** Demand is $133.5677$ units and profit is $\$2139.191$.

## (c) [4 marks]

**M1** Solve $500(p-8)e^{-0.04p} = 1200$ numerically on the given domain.

**A2** The two roots are $11.85636$ and $87.51051$ dollars.

**R1** The profit is non-negative between and including these roots.

## (d) [4 marks]

**M1** $500e^{-0.04p} \leq 100$ gives $p \geq \ln 5/0.04$.

**A1** The smallest feasible price is $40.23595$ dollars, above $33$.

**R1** The profit function is decreasing throughout the new feasible interval, so choose this boundary price.

**A1** The maximum feasible expected profit is $\$2023.595$.

## (e) [3 marks]

**R1** A change in fixed costs shifts the profit function vertically without changing its derivative.

**R1** The unconstrained maximizing price remains $33$, provided the demand and variable-cost models are unchanged.

**R1** The range of prices with non-negative profit shrinks, and could disappear if fixed costs exceed the maximum contribution before fixed costs.

Question 2 | Worked solutions

# Question 2 [Maximum mark: 18]

## (a) [3 marks]

**A1** $T = \begin{pmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{pmatrix}$

**M1** $p_{n+1} = 0.75p_n + 0.25(1 - p_n)$.

**A1** This simplifies to $0.5p_n + 0.25$.

## (b) [3 marks]

**M1** $p_{n+1} - 0.5 = 0.5(p_n - 0.5)$.

**M1** This is geometric, with initial value $p_0 - 0.5 = 0.5$.

**A1** $p_n = 0.5 + 0.5(0.5)^n$.

## (c) [2 marks]

**A1** The steady state is $(0.5, 0.5)^T$.

**R1** All entries of $T$ are positive, so the chain is regular.

## (d) [4 marks]

**M1** The possible paths are $SSR$, $SRS$, $RSS$, starting from sunny day 0.

**A1** Their probabilities are $0.75^2(0.25)$, $0.75(0.25)^2$ and $(0.25)^2(0.75)$.

**M1** Add these mutually exclusive path probabilities.

**A1** The total probability is $0.234375$.

## (e) [4 marks]

**M1** The expectation for day $n$ is $10p_n + 3(1 - p_n) = 6.5 + 3.5(0.5)^n$.

**M1** Sum over $n = 1, \ldots, 10$, using linearity of expectation even though days are dependent.

**A1** The sum is $65 + 3.5(1 - 0.5^{10})$.

**A1** The expected energy is $68.49658$ kWh.

## (f) [2 marks]

**R1** The day states are dependent: for example $P(S_{n+1} \mid S_n) = 0.75$, not $0.5$.

**R1** Also the marginal probabilities for the first three days are not all $0.5$, because the process starts sunny.

Question 3 | Worked solutions

# Question 3 [Maximum mark: 19]

## (a) [4 marks]

**M1** Use the eight raw pairs to fit $y = a + bx$.

**A1** $a = 12.83095$ and $b = -0.2095238$.

**A1** The fitted line is $y = 12.83095 + (-0.2095238)x$.

**A1** Pearson's coefficient is $r = -0.9903281$.

## (b) [5 marks]

**A1** $H_0 : \rho = 0, H_1 : \rho < 0$.

**M1** Use a correlation $t$ test on the given raw bivariate-normal pairs.

**A1** $t = r\sqrt{(8-2)/(1-r^2)} = -17.48383$, with $6$ degrees of freedom.

**A1** The lower-tail $p$-value is $1.122767 \times 10^{-6}$.

**R1** Reject $H_0$ at $1\%$; there is evidence of a negative population correlation under these assumptions.

## (c) [3 marks]

**M1** Temperatures strictly increase while oxygen values strictly decrease, giving reverse ranks without ties.

**A1** Spearman's coefficient is $-1$.

**R1** The ordering is perfectly decreasing, but the points do not lie exactly on a straight line; Pearson measures linear association rather than only rank order.

## (d) [3 marks]

**M1** Substitute into the fitted line, and take residual as observed minus predicted.

**A1** The prediction at $21^\circ\text{C}$ is $8.430952$ mg per litre.

**A1** The residual at $20^\circ\text{C}$ is $8.6 - 8.640476 = -0.04047619$ mg per litre.

Question 3 | Worked solutions

## (e) [4 marks]

**M1** Recalculate Pearson's coefficient using all nine pairs.

**A1** The new coefficient is $r = 0.246713$.

**R1** The point has strong influence, but deletion requires investigation of measurement validity and population comparability, not merely dislike of its effect.

**R1** Observational association alone does not establish causation; other site conditions may affect oxygen.

# Question 4 [Maximum mark: 19]

## (a) [3 marks]

**M1** Expected count is row total times column total divided by 180.

**A1** Every row and column total is 60, so all nine expected counts are 20.

**R1** Each expected count is greater than 5, satisfying the stated chi-square condition.

## (b) [5 marks]

**A1** $H_0$: travel mode and zone are independent; $H_1$: they are associated.

**M1** Calculate $\sum (O-E)^2/E$ over all nine cells.

**A1** $\chi^2 = 24.8$ and $df = (3-1)(3-1) = 4$.

**A1** The upper-tail $p$-value is $5.518909 \times 10^{-5}$.

**R1** Reject $H_0$ at 5%; there is evidence of an association between zone and travel mode.

## (c) [3 marks]

**M1** Compare the squared standardized deviations $(O-20)^2/20$.

**A1** Zone 3, cycle has the largest contribution, $13^2/20 = 8.45$.

**R1** Cycling is more frequent in the Zone 3 sample than expected under independence.

## (d) [4 marks]

**A1** The new rows are $(30, 20, 10)$ and $(30, 40, 50)$, with expected rows $(20, 20, 20)$ and $(40, 40, 40)$.

**M1** Use the new table and $df = (2-1)(3-1) = 2$.

**A1** $\chi^2 = 15$ and $p = 0.0005530844$.

**R1** The null is still rejected at 5%; the binary zone grouping remains associated with travel mode.

Question 4 | Worked solutions

## (e) [4 marks]

**R1** The pooled sample gives each zone equal weight because each contributes 60 observations.

**R1** The zone populations are unequal, so this need not match residents' actual distribution across zones.

**M1** Establish the estimated bus fraction within each zone.

**A1** Weight those fractions by the respective zone population proportions, rather than averaging the three fractions equally.

Question 5 | Worked solutions

# Question 5 [Maximum mark: 18]

## (a) [3 marks]

**A1** $x\prime = v$.

**A1** $v\prime = -2x - 3v$, so

$$\begin{pmatrix} x\prime \\ v\prime \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix}$$

**A1** The initial state is $(4, -2)^T$.

## (b) [4 marks]

**M1** $\det(M - \lambda I) = \lambda^2 + 3\lambda + 2$.

**A1** The eigenvalues are $-1, -2$.

**A1** Corresponding eigenvectors are $(1, -1)^T$ and $(1, -2)^T$.

**R1** The distinct negative real eigenvalues give a stable node.

## (c) [5 marks]

**M1** Write $(x, v)^T = Ae^{-t}(1, -1)^T + Be^{-2t}(1, -2)^T$.

**A1** The initial conditions give $A + B = 4$ and $-A - 2B = -2$.

**A1** Solving gives $A = 6, B = -2$.

**A1** $x(t) = 6e^{-t} - 2e^{-2t}$.

**A1** $v(t) = -6e^{-t} + 4e^{-2t}$.

## (d) [3 marks]

**R1** $x = e^{-t}(6 - 2e^{-t}) > 0$, since $0 < e^{-t} \leq 1$.

**R1** $v = e^{-t}(-6 + 4e^{-t}) < 0$ on the same domain.

**R1** Both exponentials tend to zero, so the displacement decreases toward zero without crossing it.

Question 5 | Worked solutions

## (e) [3 marks]

**M1** Use $x_{n+1} = x_n + 0.2v_n$, $v_{n+1} = v_n + 0.2(-2x_n - 3v_n)$ simultaneously. The first state is $(3.6, -2.4)$.

**A1** After the second step the Euler state is $(3.12, -2.4)$.

**A1** The exact state is $(3.123262, -2.224604)$; Euler slightly understates $x$ and makes the velocity more negative.

# Question 6 [Maximum mark: 18]

## (a) [4 marks]

**M1** The bisector of horizontal $AB$ passes through $(4, 0)$.

**A1** It is $x = 4$.

**M1** Equate $x^2 + y^2$ and $(x - 4)^2 + (y - 6)^2$.

**A1** The second bisector is $2x + 3y = 13$.

## (b) [4 marks]

**M1** Intersect $x = 4$ and $2x + 3y = 13$.

**A1** The vertex is $V = (4, 5/3)$.

**A1** The common distance is $\sqrt{16 + 25/9} = 13/3$ km.

**R1** At $x = 4$ the triangle extends from $y = 0$ to $y = 6$, so $V$ is inside.

## (c) [4 marks]

**M1** The cell is the polygon with vertices $(0, 0), (4, 0), (4, 5/3), (2, 3)$.

**A1** The final point is the midpoint of side $AC$, where its bisector meets the triangle boundary.

**M1** Use a polygon area formula or divide the region into triangles.

**A1** The area is $23/3$ km$^2$.

## (d) [4 marks]

**M1** The cell vertices consist of the posts themselves, the three side midpoints, and $V$.

**A1** Nearest distances there are $0$, $4$ at the midpoint of $AB$, $\sqrt{13}$ at the other two midpoints, and $13/3$ at $V$.

**R1** The supplied convex-cell maximum property means no interior or edge point within any cell can exceed its largest vertex distance.

**A1** The global optimum is $V = (4, 5/3)$, with nearest-post distance $13/3$ km.

## (e) [2 marks]

**R1** On $y = 0$, the nearest of $A, B$ is at most 4 km away, with equality only at $(4, 0)$; $C$ is then 6 km away.

**A1** The optimum is $(4, 0)$ and the maximum nearest-post distance is 4 km.